

WEEK 8



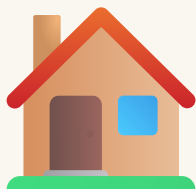
Build a Village

Final project — put it all together

마지막 프로젝트: 마을 만들기

BIG PICTURE

Everything you learned, in one build 🏠



Move, place, loop, nest, dig, variables — all together.

배운 모든 것을 한 번에

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

A village is many structures



house

walls + roof



well

wall + dig



farm house

house + farm

집 · 우물 · 농장 집

CONCEPT

The key step: move between

After each structure, the agent must **walk to a new spot** — or the next build lands on top of the last.

```
# Stage 1: house  
# move to next spot  
# Stage 2: well  
# move to next spot  
# Stage 3: farm house
```

사이마다 다음 자리로 이동해야 해요

CONCEPT

Plan with a skeleton first



skeleton

comments + stages



fill in

code each stage

먼저 뼈대를 짜고 채워 넣어요

CODE

Blocks: the village skeleton

```
# Stage 1 – house with roof  
[house walls + roof]  
move forward by 8    # to next spot  
# Stage 2 – well  
[square wall + dig down]  
move forward by 8  
# Stage 3 – farm house  
[house + farm patch]
```


CODE

How the teacher world places them

```
# each structure sits at its own x,z  
house()      # x = 10, z = 0  
well()       # x = 20, z = 0  
farm_house() # x = 10, z = 10
```

Different **x** / **z** coordinates keep each building apart — the code version of "move to the next spot".

좌표를 다르게 해서 떨어뜨려요

PREDICT 🤔

Which structure is this?

Predict first — then click reveal

```
repeat 4 times:  
  repeat 3 times:  
    place block down  
    move forward  
  turn right  
repeat 3 times:  
  destroy block down  
  move down by 1
```

Reveal output 🔍 reveal

VOCAB

New words today

village a group of buildings together

structure one finished build (house, well...)

skeleton an outline plan before the full code

project a big build made of many parts

마을

구조물

뼈대

프로젝트



What's wrong with this skeleton?

Goal: house, well, AND farm house — three buildings, side by side.

buggy

```
# House
[house walls + roof]
# Well
[well code]
```

Think first — then click next.

**DEBUG**

The bug

Bug: the farm house is **missing**, and there's no "move to next spot" between buildings — the well would build on top of the house.

fixed

```
# House
[house walls + roof]
move to next spot
# Well
[well code]
move to next spot
# Farm house
[farm house code]
```

YOUR TURN!

Build your village



Open this week's world



Build a **house with a roof**



Move over, build a **well**



Move over, build a **farm house**



Send a video walking through your village



QUIZ

QUESTION 1 OF 3

Why move the agent between structures?

So the next build doesn't land on the last one.

To delete the village.

It isn't needed.

To make the code shorter.



QUIZ

QUESTION 2 OF 3

What is a code **skeleton**?

An outline of the stages before the full code.

A finished, working program.

A type of Minecraft mob.

A single loop.



QUIZ

QUESTION 3 OF 3

A square wall plus digging straight down makes a:

Farm patch.

Roof.

Well.

Bridge.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Village complete!

You finished Builder Basics. 🎉

빌더 기초 과정 완주 — 정말 잘했어요!